Q. Implement a Random Forest classifier to predict customer churn in a telecom dataset and evaluate its performance using accuracy, confusion matrix, and classification report metrics.

Ans:

MODEL EVALUATION RESULTS

Accuracy Score: 0.7963 (79.63%)

Confusion Matrix: [[945 91],

[196 177]]

Interpretation:

True Negatives (Correctly predicted Retained): 945

False Positives (Incorrectly predicted Churned): 91

False Negatives (Incorrectly predicted Retained): 196

True Positives (Correctly predicted Churned): 177

Classification Report:

	precision	recall	f1-score	support
Retained (0)	0.83	0.91	0.87	1036
Churned (1)	0.66	0.47	0.55	373
accuracy			0.80	1409
macro avg	0.74	0.69	0.71	1409
weighted avg	0.78	0.80	0.78	1409

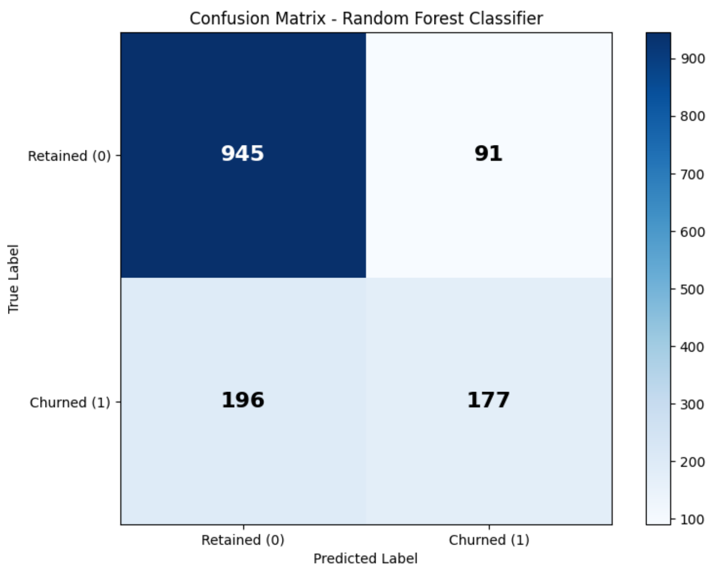

Fig: Confusion Matrix